Title

A Data-Driven Analysis of Crimes Against Women in India (2001–2021): Patterns, Correlations, and Trends

1.Introduction

Crimes against women remain one of the most pressing social challenges in India. Understanding the patterns and evolution of such crimes is essential for designing effective interventions and policies.

This project performs an Exploratory Data Analysis (EDA) on crime data from 2001 to 2021, aiming to uncover hidden patterns, identify high-risk regions, and analyse how different types of crimes are interrelated over time.

2.Objective

The primary objective of this study is to:

• Analyse the distribution of crimes across different states

• Identify the most prevalent types of crimes against women

• Examine year-wise trends and growth patterns

• Understand relationships between different crime categories

• Extract meaningful insights that can support awareness and decision-making

3.Dataset Description

The dataset consists of crime records against women across Indian states from 2001 to 2021.

Key attributes include:

- State – Region where crime was reported
- Year – Time of occurrence
- Rape – Cases of sexual assault
- Kidnapping & Abduction (K&A)
- Dowry Death (DD)
- Assault on Women (AoW)
- Insult to Modesty (AoM)
- Domestic Violence (DV)
- Women Trafficking (WT)

The dataset contains 736 records with no missing values after preprocessing.

4.Data Preprocessing

Before analysis, the dataset was cleaned and prepared:

- Removed unnecessary index column (Unnamed: 0)
- Checked and confirmed absence of missing values

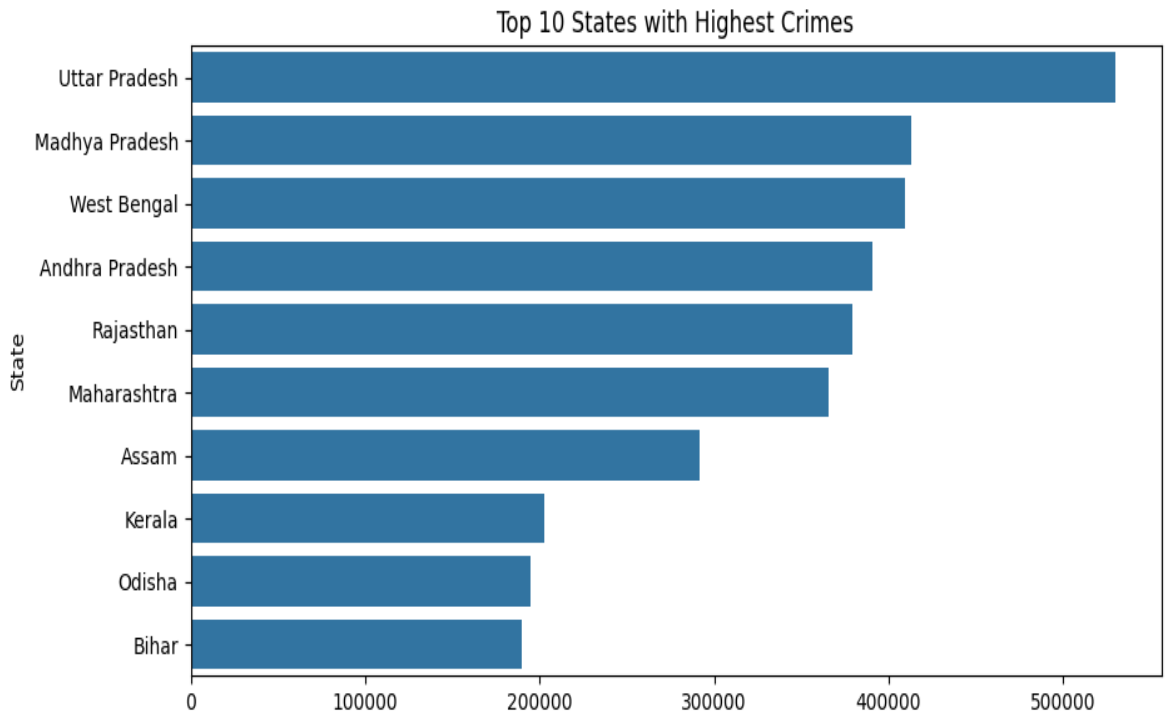
• Standardized column names and formats

• Created a new feature: Total Crime (sum of all crime categories)

This ensured consistency and reliability for further analysis.

5.Exploratory Data Analysis

5.1 State-wise Crime Distribution

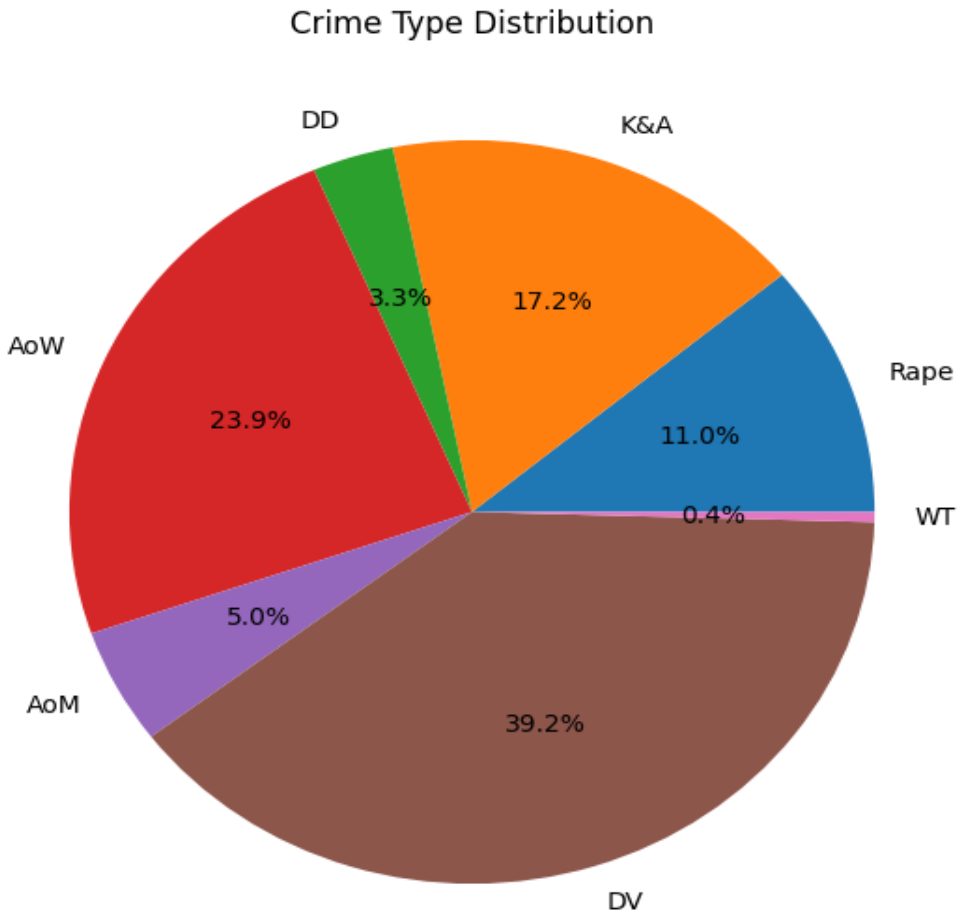


A bar plot was used to identify the top 10 states with the highest crime counts.

Observation:

States like Uttar Pradesh, Madhya Pradesh, and West Bengal consistently show higher crime rates, indicating potential socio-economic and population-related influences.

5.2 Crime Type Contribution

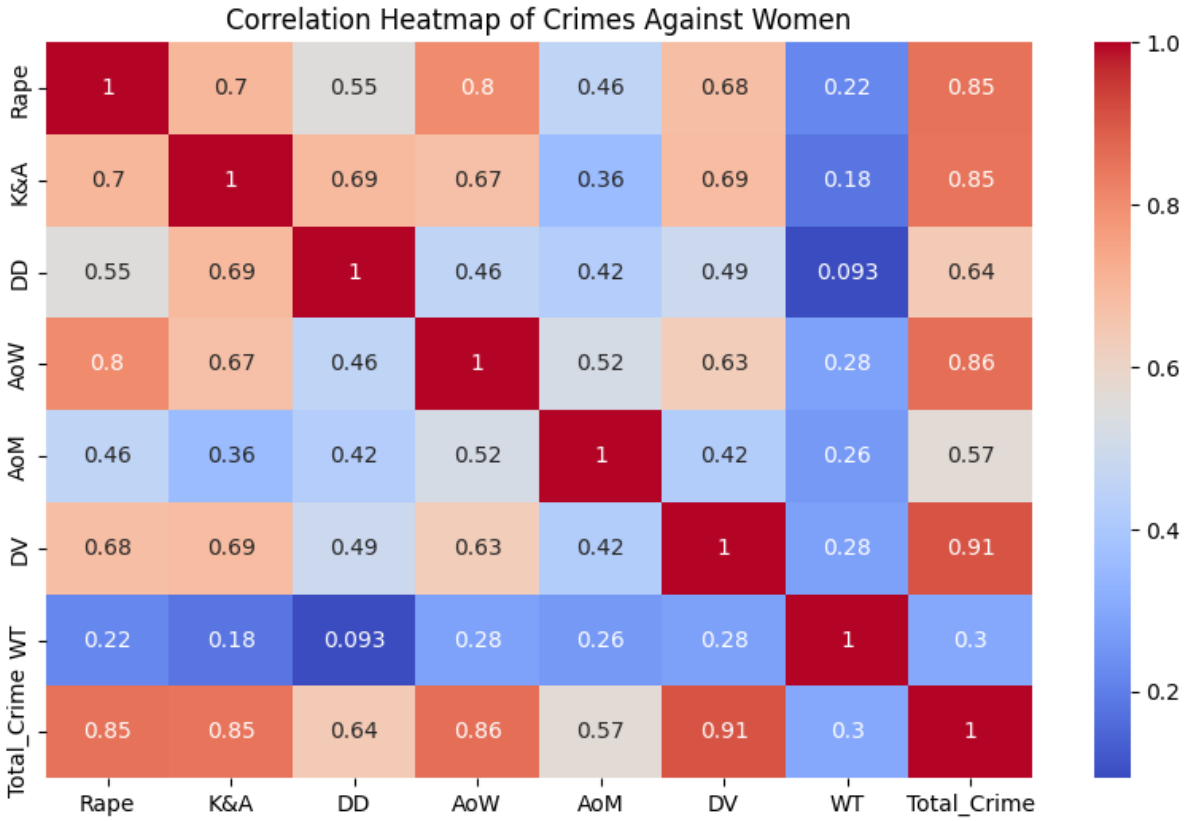


A pie chart was used to analyse the proportion of each crime type.

Observation:

Domestic Violence accounts for the largest share of crimes, followed by Assault on Women and Kidnapping & Abduction. Women Trafficking contributes the least.

5.3 Correlation Analysis



A heatmap was generated to examine relationships between different crime types.

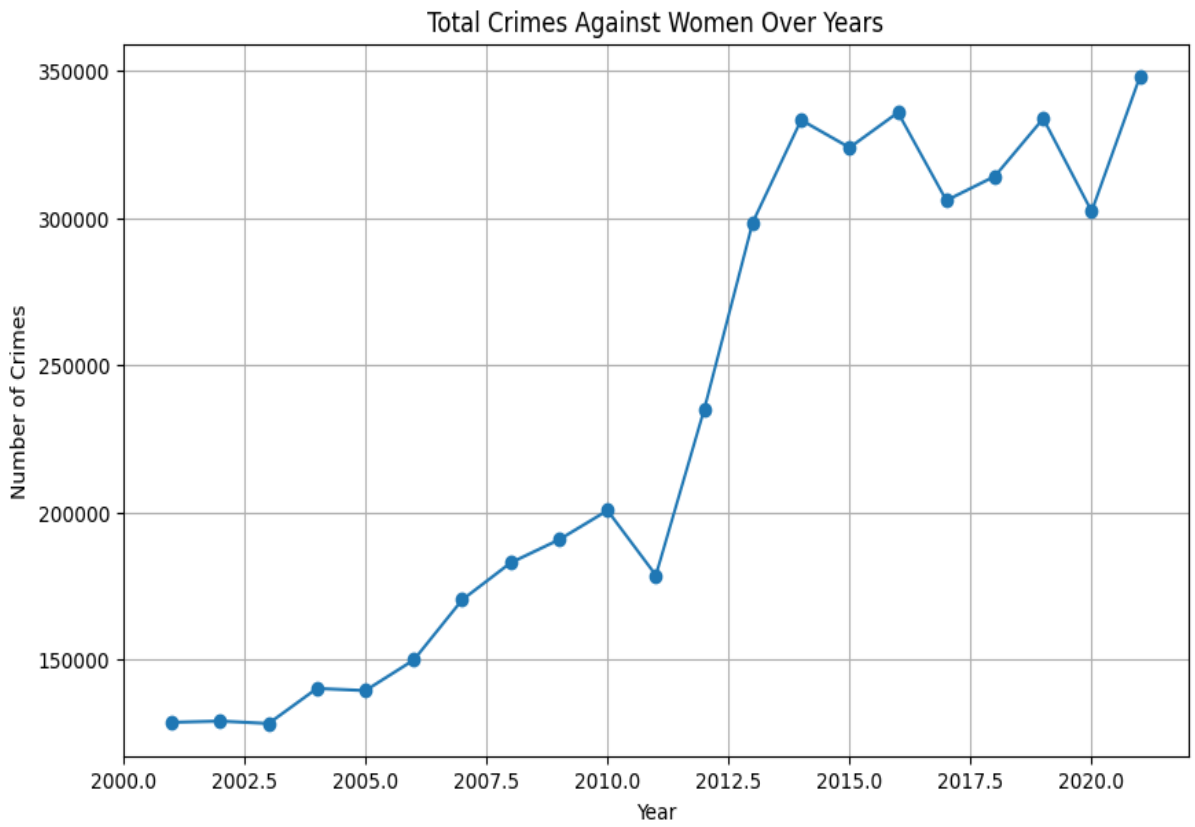
Observation:

• Strong positive correlation between Domestic Violence and Total Crime

• Moderate correlations among multiple crime categories

• Indicates that states with high crime in one category often show high crime in others

5.4 Year-wise Trend Analysis



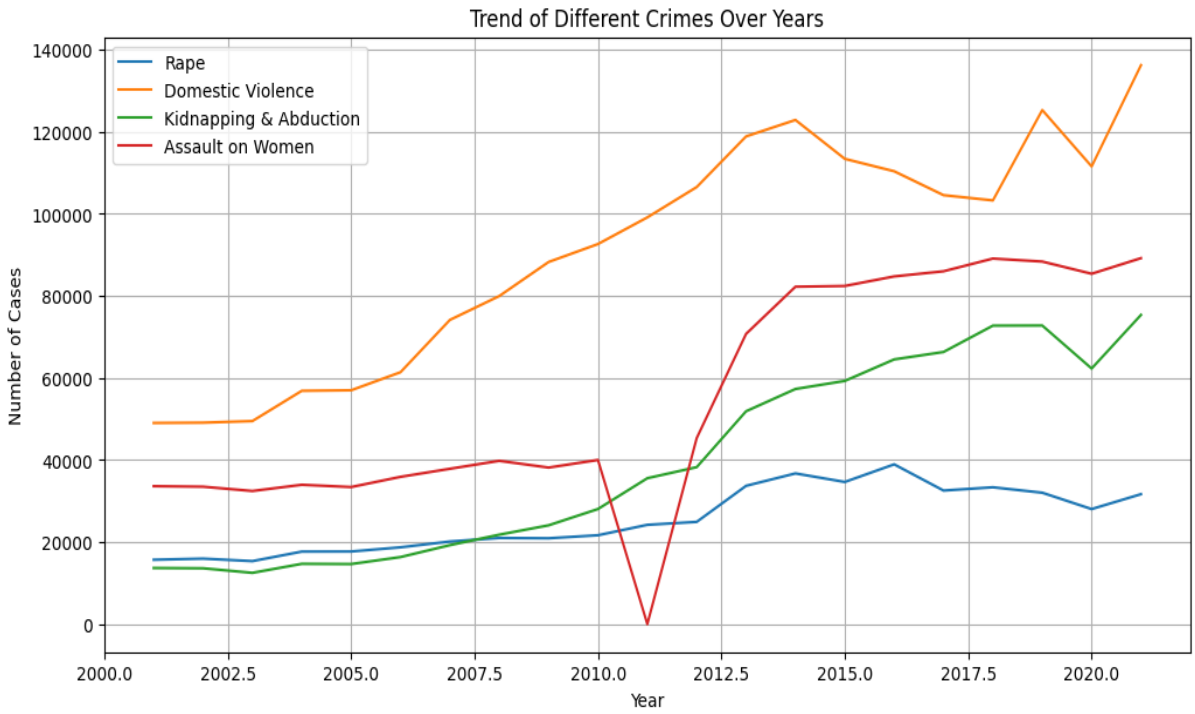
A line graph was used to track total crimes over time.

Observation:

- Gradual increase from 2001 to 2010
- Sharp rise after 2012
- Slight fluctuations after 2016 but overall upward trend

This suggests increased reporting, awareness, or actual rise in incidents.

5.5 Crime-wise Trend Comparison



Multiple line plots were used to compare different crime categories over time.

Observation:

• Domestic Violence shows a steady and dominant increase

• Kidnapping & Abduction also shows significant growth

• Rape cases increase gradually but remain comparatively lower than DV

• Sudden anomaly observed in Assault on Women around 2011 (possible reporting inconsistency)

6. Key Insights

• Domestic Violence is the most dominant crime category

• Uttar Pradesh consistently ranks highest in crime count

• Crimes show a noticeable surge after 2012

• Multiple crime categories are strongly interlinked

• Urbanization, awareness, and reporting mechanisms may influence trends

7. Limitations

• Data does not include population normalization (per capita analysis missing)

• Possible underreporting in earlier years

• No socio-economic factors included for deeper analysis

8.Real World Implication

The findings of this analysis have significant real-world implications.

The concentration of crimes in certain states indicates the need for region-specific policy interventions. The dominance of domestic violence highlights the importance of strengthening household-level awareness and support systems.

The sharp rise in crimes after 2012 may reflect increased reporting due to legal reforms and public awareness, suggesting

that data trends are influenced not only by incidents but also by societal response.

Such data-driven insights can assist government bodies, NGOs, and policymakers in designing targeted safety measures and awareness campaigns.

9. Conclusion

This analysis highlights critical patterns in crimes against women across India. The increasing trend over the years emphasizes the need for stronger policies, awareness programs, and enforcement mechanisms.

While the dataset provides valuable insights, further analysis incorporating demographic and socio-economic factors can enhance understanding and lead to more targeted solutions.